

AERIS Expert Review Packet

Anomaly

Source / metric	openaq / Ozone (ozone)
Observed / expected baseline	0.101 / 0.0185629 ppm
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-04 23:00 UTC
Location	29.7681, -95.2206
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 869	36.74 to 100; mean 72.1848	59.35 at 2026-07-04 22:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 869	22.7778 to 36.6667; mean 29.5219	33 at 2026-07-04 22:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 1166	0 to 360; mean 143.756	170 at 2026-07-04 22:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 1201	0 to 9.77444; mean 2.2445	8.2311 at 2026-07-04 22:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	9 / 108	5903.97 to 5930.15; mean 5918.46	5916.99 at 2026-07-05 00:00Z; 1 h after event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	9 / 108	11.4652 to 2081.56; mean 769.689	1199.55 at 2026-07-05 00:00Z; 1 h after event
noaa_gfs	Precipitable water (precipitable_water)	mm	9 / 108	39.7061 to 52.0889; mean 45.1575	50.5099 at 2026-07-05 00:00Z; 1 h after event
noaa_gfs	Surface pressure (surface_pressure)	hPa	9 / 108	1006.56 to 1016.78; mean 1012.86	1011.87 at 2026-07-05 00:00Z; 1 h after event
noaa_gfs	850-hPa temperature (t_850)	degC	9 / 108	17.3943 to 20.6923; mean 19.1897	19.6205 at 2026-07-05 00:00Z; 1 h after event
noaa_gfs	10-m eastward wind (u_10m)	m/s	9 / 108	-2.73237 to 3.62264; mean 0.3918	-1.43237 at 2026-07-05 00:00Z; 1 h after event
noaa_gfs	10-m northward wind (v_10m)	m/s	9 / 108	0.184915 to 5.71323; mean 2.1941	4.76577 at 2026-07-05 00:00Z; 1 h after event
openaq	Ozone (ozone)	ppm	17 / 1127	-0.004 to 0.101; mean 0.0174	0.101 at 2026-07-04 23:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 218	10 to 61; mean 20.8578	24 at 2026-07-04 23:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	81 / 3660	0 to 94.9; mean 6.8687	6.3 at 2026-07-04 23:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 359	0 to 100; mean 45.9833	42 at 2026-07-04 23:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 359	50 to 95; mean 74.9443	70 at 2026-07-04 23:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 359	0 to 1.85; mean 0.006	0 at 2026-07-04 23:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 359	1011 to 1018; mean 1014.99	1013 at 2026-07-04 23:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 359	24.58 to 36.43; mean 30.2326	33.28 at 2026-07-04 23:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 359	0 to 340; mean 189.691	160 at 2026-07-04 23:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 359	0.45 to 6.09; mean 2.0743	5.26 at 2026-07-04 23:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	67 / 4804	0 to 372.3; mean 8.018	6.7 at 2026-07-04 22:59Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-04 19:52Z; 3 h 7 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-04 19:52Z; 3 h 7 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	6 / 6	0.0230724 to 0.0260625; mean 0.0248	0.0251122 at 2026-07-04 19:52Z; 3 h 7 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-07-04 19:52Z; 3 h 7 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	0.000130424 to 0.000227449; mean 0.0002	0.000130424 at 2026-07-04 19:52Z; 3 h 7 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-04 19:52Z; 3 h 7 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	4 / 4	1.96909e-05 to 2.66684e-05; mean 0	2.30269e-05 at 2026-07-04 19:52Z; 3 h 7 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-07-04 19:52Z; 3 h 7 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-04 19:52Z; 3 h 7 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-4.80456e-05 to 3.99704e-06; mean -0	1.54923e-06 at 2026-07-04 19:52Z; 3 h 7 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-04 19:52Z; 3 h 7 min before event
tceq	Carbon monoxide (co)	ppm	3 / 185	0.1 to 0.8; mean 0.2308	0.2 at 2026-07-04 23:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 746	-2.3 to 20.3; mean 3.3587	1.7 at 2026-07-04 23:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 182	-0.6 to 19.3; mean 0.2489	0 at 2026-07-04 23:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-03 06Z 93.35, 12Z 68.39, 18Z 62.53, 07-04 00Z 76.09, 06Z 92.1, 12Z 67.57, 18Z 51.93, 07-05 00Z 75.63, 06Z 91.22, 12Z 69.81, 18Z 47.96, 07-06 00Z 73.47, 06Z 87.79
asos	Air temperature (temperature)	07-03 06Z 24.53, 12Z 30.5, 18Z 31.49, 07-04 00Z 28.66, 06Z 25.6, 12Z 30.58, 18Z 33.91, 07-05 00Z 28.65, 06Z 25.68, 12Z 30.5, 18Z 34.07, 07-06 00Z 28.88, 06Z 26.18
asos	Wind direction (wind_direction)	07-03 06Z 1.176, 12Z 105.3, 18Z 154.5, 07-04 00Z 160.1, 06Z 65.94, 12Z 189.3, 18Z 142, 07-05 00Z 164.4, 06Z 79.15, 12Z 201.8, 18Z 203.9, 07-06 00Z 153.8, 06Z 130.2
asos	Wind speed (wind_speed)	07-03 06Z 0.09078, 12Z 1.421, 18Z 3.509, 07-04 00Z 2.735, 06Z 0.7313, 12Z 2.202, 18Z 3.224, 07-05 00Z 3.081, 06Z 0.7639, 12Z 2.378, 18Z 2.71, 07-06 00Z 3.066, 06Z 1.347
noaa_gfs	500-hPa geopotential height (gh_500)	07-03 12Z 5915, 18Z 5928, 07-04 00Z 5926, 06Z 5929, 12Z 5919, 18Z 5930, 07-05 00Z 5917, 06Z 5922, 12Z 5907, 18Z 5916, 07-06 00Z 5905, 06Z 5907
noaa_gfs	Planetary boundary layer height (pbl_height)	07-03 12Z 78.84, 18Z 1697, 07-04 00Z 801.5, 06Z 288.8, 12Z 122.3, 18Z 1879, 07-05 00Z 911.2, 06Z 295.7, 12Z 173.3, 18Z 1808, 07-06 00Z 907.4, 06Z 272.5
noaa_gfs	Precipitable water (precipitable_water)	07-03 12Z 42.13, 18Z 47.33, 07-04 00Z 48.85, 06Z 45.55, 12Z 42.3, 18Z 45.85, 07-05 00Z 49.69, 06Z 43.79, 12Z 43.14, 18Z 43.52, 07-06 00Z 44.23, 06Z 45.52
noaa_gfs	Surface pressure (surface_pressure)	07-03 12Z 1015, 18Z 1015, 07-04 00Z 1013, 06Z 1014, 12Z 1014, 18Z 1014, 07-05 00Z 1011, 06Z 1013, 12Z 1013, 18Z 1013, 07-06 00Z 1009, 06Z 1012
noaa_gfs	850-hPa temperature (t_850)	07-03 12Z 18.61, 18Z 17.87, 07-04 00Z 19.54, 06Z 19.63, 12Z 19.5, 18Z 18.22, 07-05 00Z 19.75, 06Z 19.72, 12Z 19.3, 18Z 18.04, 07-06 00Z 20.14, 06Z 19.97
noaa_gfs	10-m eastward wind (u_10m)	07-03 12Z 0.2824, 18Z -0.1871, 07-04 00Z -1.516, 06Z 1.028, 12Z 0.9244, 18Z 0.5623, 07-05 00Z -1.421, 06Z 1.478, 12Z 1.609, 18Z 0.7035, 07-06 00Z -0.1885, 06Z 1.427
noaa_gfs	10-m northward wind (v_10m)	07-03 12Z 0.7738, 18Z 1.827, 07-04 00Z 3.432, 06Z 2.48, 12Z 0.9778, 18Z 1.705, 07-05 00Z 4.155, 06Z 2.37, 12Z 1.024, 18Z 1.396, 07-06 00Z 3.735, 06Z 2.454
openaq	Ozone (ozone)	07-03 06Z 0.002824, 12Z 0.01309, 18Z 0.02889, 07-04 00Z 0.01811, 06Z 0.00689, 12Z 0.01455, 18Z 0.03303, 07-05 00Z 0.01641, 06Z 0.006936, 12Z 0.01391, 18Z 0.0299, 07-06 00Z 0.02046, 06Z 0.006651
openaq	PM10 (pm10)	07-03 06Z 35, 12Z 35, 18Z 26.28, 07-04 00Z 20.61, 06Z 18.94, 12Z 16.83, 18Z 17.28, 07-05 00Z 22.39, 06Z 24.78, 12Z 18.72, 18Z 16.59, 07-06 00Z 15.89, 06Z 14.39
openaq	PM2.5 (pm25)	07-03 06Z 6.739, 12Z 6.066, 18Z 6.436, 07-04 00Z 3.979, 06Z 5.607, 12Z 5.049, 18Z 5.314, 07-05 00Z 12.2, 06Z 15.77, 12Z 6.563, 18Z 4.781, 07-06 00Z 5.329, 06Z 6.732
openweather	Cloud cover (cloud_cover)	07-03 06Z 2.2, 12Z 3.034, 18Z 58.97, 07-04 00Z 18.87, 06Z 6.067, 12Z 2.448, 18Z 38.74, 07-05 00Z 89.97, 06Z 89.03, 12Z 28.23, 18Z 92.41, 07-06 00Z 88.13, 06Z 40.5
openweather	Relative humidity (humidity)	07-03 06Z 91.8, 12Z 77.62, 18Z 64.83, 07-04 00Z 74.19, 06Z 89.03, 12Z 77.9, 18Z 58.42, 07-05 00Z 74.5, 06Z 87.73, 12Z 78.83, 18Z 57.69, 07-06 00Z 72.74, 06Z 85.38
openweather	Precipitation rate (precipitation)	07-03 06Z 0, 12Z 0, 18Z 0.065, 07-04 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-05 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-06 00Z 0.006774, 06Z 0
openweather	Surface pressure (pressure)	07-03 06Z 1016, 12Z 1018, 18Z 1016, 07-04 00Z 1015, 06Z 1016, 12Z 1017, 18Z 1014, 07-05 00Z 1014, 06Z 1015, 12Z 1016, 18Z 1013, 07-06 00Z 1013, 06Z 1014
openweather	Air temperature (temperature)	07-03 06Z 24.87, 12Z 29.81, 18Z 33.16, 07-04 00Z 30.08, 06Z 26.53, 12Z 29.88, 18Z 35.03, 07-05 00Z 30.05, 06Z 26.68, 12Z 29.88, 18Z 34.99, 07-06 00Z 30.2, 06Z 26.66
openweather	Wind direction (wind_direction)	07-03 06Z 188.2, 12Z 191.9, 18Z 163.4, 07-04 00Z 164.8, 06Z 189, 12Z 223.3, 18Z 171.7, 07-05 00Z 169.9, 06Z 194.7, 12Z 216.4, 18Z 196.2, 07-06 00Z 175.9, 06Z 230.8
openweather	Wind speed (wind_speed)	07-03 06Z 1.274, 12Z 1.343, 18Z 2.034, 07-04 00Z 2.224, 06Z 1.713, 12Z 1.607, 18Z 2.431, 07-05 00Z 2.49, 06Z 1.986, 12Z 1.85, 18Z 2.166, 07-06 00Z 2.95, 06Z 2.165

Source	Metric	Values
purpleair	PM2.5 (pm25)	07-03 06Z 9.151, 12Z 6.815, 18Z 4.813, 07-04 00Z 5.359, 06Z 6.96, 12Z 6.325, 18Z 6.014, 07-05 00Z 14.81, 06Z 15.54, 12Z 6.754, 18Z 5.607, 07-06 00Z 8.65, 06Z 8.72
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-03 18Z 1, 07-04 18Z 1, 07-05 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-03 18Z 1, 07-04 18Z 1, 07-05 18Z 1
sentinel5p	CO column (s5p_co_column)	07-03 18Z 0.02605, 07-04 18Z 0.02512, 07-05 18Z 0.02308
sentinel5p	CO granules available (s5p_co_granule_available)	07-03 18Z 1, 07-04 18Z 1, 07-05 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	07-03 18Z 0.0002216, 07-04 18Z 0.0001321, 07-05 18Z 0.0002018
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-03 18Z 1, 07-04 18Z 1, 07-05 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	07-03 18Z 2.417e-05, 07-04 18Z 2.303e-05, 07-05 18Z 1.969e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-03 18Z 1, 07-04 18Z 1, 07-05 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-03 18Z 1, 07-04 18Z 1, 07-05 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	07-03 18Z -3.098e-05, 07-04 18Z 2.773e-06, 07-05 18Z -4.805e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-03 18Z 1, 07-04 18Z 1, 07-05 18Z 1
tceq	Carbon monoxide (co)	07-03 06Z 0.4333, 12Z 0.2389, 18Z 0.3, 07-04 00Z 0.2667, 06Z 0.2111, 12Z 0.2167, 18Z 0.2222, 07-05 00Z 0.24, 06Z 0.1944, 12Z 0.2, 18Z 0.2071, 07-06 00Z 0.21, 06Z 0.25
tceq	Nitrogen dioxide (no2)	07-03 06Z 7.417, 12Z 4.446, 18Z 3.808, 07-04 00Z 3.593, 06Z 4.051, 12Z 2.953, 18Z 2.513, 07-05 00Z 1.951, 06Z 3.896, 12Z 2.649, 18Z 2.067, 07-06 00Z 2.523, 06Z 4.436
tceq	Sulfur dioxide (so2)	07-03 06Z -0.2, 12Z -0.1167, 18Z -0.1353, 07-04 00Z -0.1667, 06Z -0.2222, 12Z 0.02778, 18Z 0.5444, 07-05 00Z -0.1333, 06Z -0.2437, 12Z 0.1706, 18Z 3.087, 07-06 00Z 0.11, 06Z -0.2

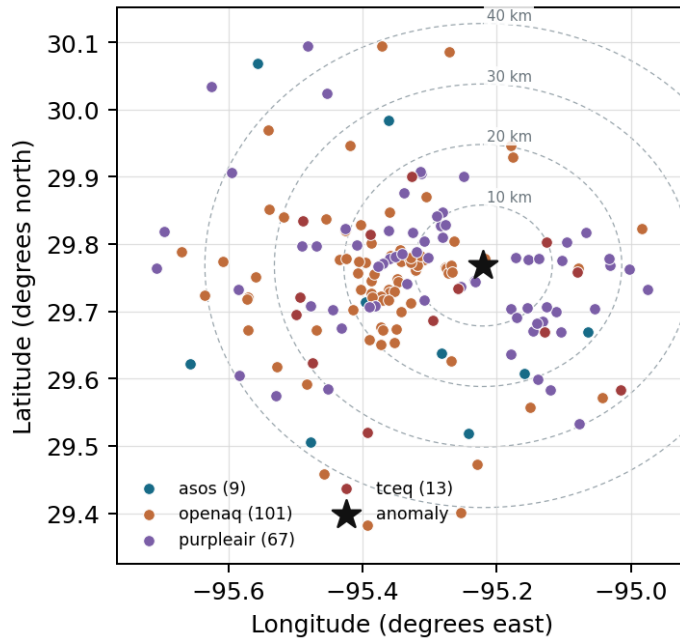
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

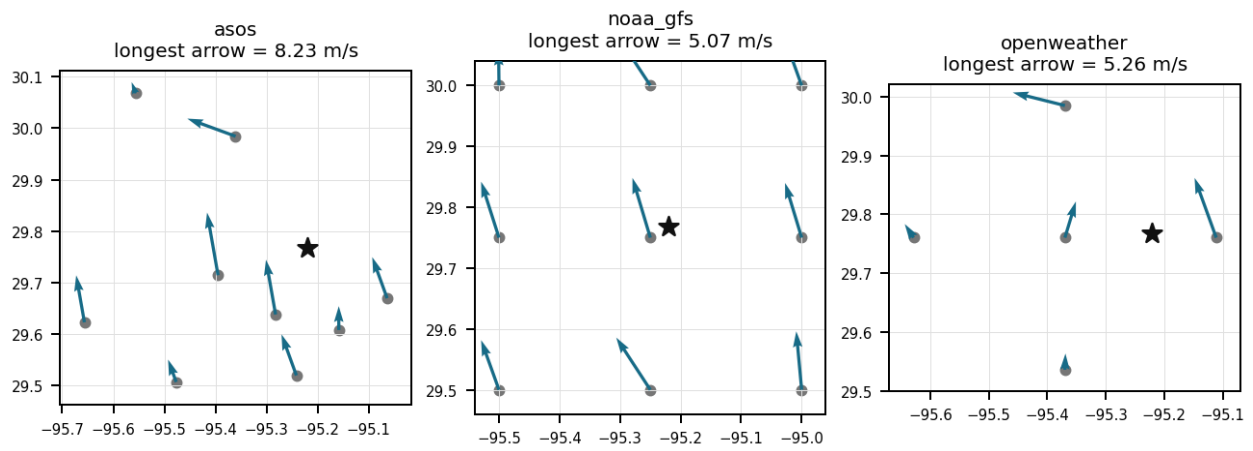
Source	Metric	Values
asos	Relative humidity (humidity)	HOU (15.701 km) 71.19, MCJ (17.88 km) 70.48, T41 (18.691 km) 72.16, EFD (18.853 km) 73.22, IAH (27.585 km) 69.4, LVJ (27.785 km) 72.85, AXH (38.24 km) 71.91, SGR (45.118 km) 75.44, +1 more
asos	Air temperature (temperature)	HOU (15.701 km) 30.17, MCJ (17.88 km) 29.82, T41 (18.691 km) 29.68, EFD (18.853 km) 29.58, IAH (27.585 km) 30.09, LVJ (27.785 km) 29.29, AXH (38.24 km) 28.96, SGR (45.118 km) 29.4, +1 more
asos	Wind direction (wind_direction)	HOU (15.701 km) 166.5, MCJ (17.88 km) 138.3, T41 (18.691 km) 189, EFD (18.853 km) 145.8, IAH (27.585 km) 158.6, LVJ (27.785 km) 117.4, AXH (38.24 km) 81.14, SGR (45.118 km) 157.8, +1 more
asos	Wind speed (wind_speed)	HOU (15.701 km) 2.769, MCJ (17.88 km) 2.959, T41 (18.691 km) 2.86, EFD (18.853 km) 2.63, IAH (27.585 km) 2.326, LVJ (27.785 km) 1.715, AXH (38.24 km) 0.9995, SGR (45.118 km) 2.471, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (3.479 km) 5919, gfs:29.75,-95.00 (21.39 km) 5918, gfs:30.00,-95.25 (25.941 km) 5918, gfs:29.75,-95.50 (27.046 km) 5919, gfs:29.50,-95.25 (29.946 km) 5919, gfs:30.00,-95.00 (33.425 km) 5918, gfs:29.50,-95.00 (36.651 km) 5919, gfs:30.00,-95.50 (37.29 km) 5918, +1 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (3.479 km) 891.5, gfs:29.75,-95.00 (21.39 km) 755.6, gfs:30.00,-95.25 (25.941 km) 964.7, gfs:29.75,-95.50 (27.046 km) 989.8, gfs:29.50,-95.25 (29.946 km) 579.9, gfs:30.00,-95.00 (33.425 km) 552.9, gfs:29.50,-95.00 (36.651 km) 636.9, gfs:30.00,-95.50 (37.29 km) 975.4, +1 more

Source	Metric	Values
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (3.479 km) 45.45, gfs:29.75,-95.00 (21.39 km) 45.45, gfs:30.00,-95.25 (25.941 km) 45.24, gfs:29.75,-95.50 (27.046 km) 44.87, gfs:29.50,-95.25 (29.946 km) 45.5, gfs:30.00,-95.00 (33.425 km) 45.12, gfs:29.50,-95.00 (36.651 km) 45.14, gfs:30.00,-95.50 (37.29 km) 44.62, +1 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (3.479 km) 1013, gfs:29.75,-95.00 (21.39 km) 1014, gfs:30.00,-95.25 (25.941 km) 1012, gfs:29.75,-95.50 (27.046 km) 1012, gfs:29.50,-95.25 (29.946 km) 1014, gfs:30.00,-95.00 (33.425 km) 1013, gfs:29.50,-95.00 (36.651 km) 1015, gfs:30.00,-95.50 (37.29 km) 1010, +1 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (3.479 km) 19.19, gfs:29.75,-95.00 (21.39 km) 19.21, gfs:30.00,-95.25 (25.941 km) 19.17, gfs:29.75,-95.50 (27.046 km) 19.16, gfs:29.50,-95.25 (29.946 km) 19.19, gfs:30.00,-95.00 (33.425 km) 19.07, gfs:29.50,-95.00 (36.651 km) 19.3, gfs:30.00,-95.50 (37.29 km) 19.21, +1 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (3.479 km) 0.552, gfs:29.75,-95.00 (21.39 km) 0.1187, gfs:30.00,-95.25 (25.941 km) 0.2353, gfs:29.75,-95.50 (27.046 km) 0.4937, gfs:29.50,-95.25 (29.946 km) 0.2853, gfs:30.00,-95.00 (33.425 km) 0.302, gfs:29.50,-95.00 (36.651 km) 0.2853, gfs:30.00,-95.50 (37.29 km) 1.094, +1 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (3.479 km) 2.355, gfs:29.75,-95.00 (21.39 km) 2.189, gfs:30.00,-95.25 (25.941 km) 1.947, gfs:29.75,-95.50 (27.046 km) 2.314, gfs:29.50,-95.25 (29.946 km) 2.255, gfs:30.00,-95.00 (33.425 km) 2.064, gfs:29.50,-95.00 (36.651 km) 2.672, gfs:30.00,-95.50 (37.29 km) 1.614, +1 more
openaq	Ozone (ozone)	2039 (0.0 km) 0.01867, 3378 (5.211 km) 0.000125, 2046 (9.065 km) 0.02076, 3214 (9.934 km) 0.0206, 325 (11.566 km) 0.004941, 2800 (13.761 km) 0.0195, 332 (14.072 km) 0.01941, 320 (16.472 km) 0.01304, +9 more
openaq	PM10 (pm10)	13380082 (5.211 km) 19.92, 15852419 (9.388 km) 24.96, 271 (14.072 km) 17.65
openaq	PM2.5 (pm25)	13787195 (1.131 km) 7.749, 9201803 (4.501 km) 7.608, 9489522 (4.603 km) 6.813, 8967065 (5.194 km) 6.935, 3379 (5.211 km) 7.316, 15539682 (5.23 km) 7.891, 8967123 (5.244 km) 8.247, 8967479 (5.244 km) 7.471, +73 more
openweather	Cloud cover (cloud_cover)	grid:east (10.604 km) 42.99, grid:center (14.427 km) 47.28, grid:north (28.083 km) 45.29, grid:south (29.582 km) 46.9, grid:west (39.382 km) 47.47
openweather	Relative humidity (humidity)	grid:east (10.604 km) 79.26, grid:center (14.427 km) 73.49, grid:north (28.083 km) 73.62, grid:south (29.582 km) 74.46, grid:west (39.382 km) 73.86
openweather	Precipitation rate (precipitation)	grid:east (10.604 km) 0.02569, grid:center (14.427 km) 0, grid:north (28.083 km) 0.004306, grid:south (29.582 km) 0, grid:west (39.382 km) 0
openweather	Surface pressure (pressure)	grid:east (10.604 km) 1015, grid:center (14.427 km) 1015, grid:north (28.083 km) 1015, grid:south (29.582 km) 1015, grid:west (39.382 km) 1015
openweather	Air temperature (temperature)	grid:east (10.604 km) 29.84, grid:center (14.427 km) 30.78, grid:north (28.083 km) 30.27, grid:south (29.582 km) 30.02, grid:west (39.382 km) 30.26
openweather	Wind direction (wind_direction)	grid:east (10.604 km) 189.8, grid:center (14.427 km) 205.7, grid:north (28.083 km) 179.1, grid:south (29.582 km) 190.6, grid:west (39.382 km) 183.5
openweather	Wind speed (wind_speed)	grid:east (10.604 km) 2.786, grid:center (14.427 km) 1.82, grid:north (28.083 km) 1.898, grid:south (29.582 km) 2.173, grid:west (39.382 km) 1.69
purpleair	PM2.5 (pm25)	166435 (2.815 km) 10.37, 97279 (2.97 km) 9.756, 99797 (4.634 km) 8.079, 243843 (4.892 km) 12.74, 166441 (6.532 km) 7.344, 127441 (7.393 km) 14.21, 222601 (7.938 km) 4.256, 186271 (8.063 km) 9.329, +59 more
tceq	Carbon monoxide (co)	482011035 (5.231 km) 0.2031, 482011039 (14.096 km) 0.1298, 482011052 (16.937 km) 0.3484
tceq	Nitrogen dioxide (no2)	482011035 (5.231 km) 4.04, 482010026 (9.95 km) 4.669, 482010416 (11.575 km) 3.889, 482011015 (13.673 km) 4.744, 482011039 (14.096 km) 2.037, 482011052 (16.937 km) 6.924, 482010024 (17.947 km) 1.766, 482011066 (26.769 km) 3.597, +4 more
tceq	Sulfur dioxide (so2)	482011035 (5.231 km) 0.009375, 482011039 (14.096 km) -0.05091, 482010051 (29.284 km) 0.754

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 35

At the time of the ozone anomaly on July 4, 2026, near-surface meteorological observations recorded elevated temperatures reaching 33.0 °C to 35.03 °C alongside southerly winds.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 2 of 35

The PM2.5 levels are elevated at 6.7 ug/m3, which is higher than the mean of 8.018 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 3 of 35

Light southerly surface winds and a declining late-afternoon planetary boundary layer restricted atmospheric dispersion, concentrating ozone near the surface around 23:00 UTC.

Mark exactly one:

☐ V

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Note (optional):

Claim 4 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:

☐ V

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☐ U

Note (optional):

Claim 5 of 35

Ground-based ambient monitors and satellite column observations confirmed the presence of nitrogen dioxide and formaldehyde precursors necessary to fuel localized photochemical ozone production.

Mark exactly one:

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Note (optional):

Claim 6 of 35

The openaq ozone dataset within 50 km from 2026-07-03T11:00:00+00:00 to 2026-07-06T11:00:00+00:00 had a mean of 0.0174 ppm and a maximum of 0.101 ppm, with the maximum occurring at the anomalous sensor reading.

Mark exactly one:

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Note (optional):

Claim 7 of 35

A decrease in planetary boundary layer height from approximately 1879 meters around 18:00 UTC to 911 meters by 00:00 UTC trapped photochemically generated ozone near the surface, exacerbating local ground-level concentrations.

Mark exactly one:

☐ V

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Note (optional):

Claim 8 of 35

A severe thunderstorm or tornado event in the Houston area has stirred up dust and debris, contributing to the air quality anomaly.

Mark exactly one:

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Note (optional):

Claim 9 of 35

Hot afternoon temperatures near 33 to 35 degC, relatively low afternoon humidity, no precipitation, partial sunshine, and a deep daytime boundary layer favored strong photochemical ozone production in Houston at the time of the anomaly.

Mark exactly one:

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Note (optional):

Claim 10 of 35

Southerly to south-southeasterly winds of roughly 5 to 8 m/s were capable of transporting an aged ozone-rich urban-industrial plume toward the eastern Houston anomaly site during late afternoon on 2026-07-04.

Mark exactly one:

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Note (optional):

Claim 11 of 35

On July 4, 2026 at 23:00 UTC, surface ozone concentrations at the Houston sensor spiked to an anomalous high of 0.101 ppm compared to an expected baseline of 0.0186 ppm.

Mark exactly one:

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Note (optional):

Claim 12 of 35

High ambient temperatures reaching over 36 degC observed by surface weather networks on July 4 provided strong thermal and photochemical conditions conducive to rapid atmospheric ozone formation in the Houston region.

Mark exactly one:

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Note (optional):

Claim 13 of 35

The planetary boundary layer height contracted from approximately 1879 meters at 18:00 UTC to around 911 meters by 00:00 UTC, trapping ozone near the surface.

Mark exactly one:

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Note (optional):

Claim 14 of 35

Regional meteorology in eastern Houston at 2026-07-04 23:00 UTC strongly supported photochemical ozone production because surface temperatures were near 33 to 35 C, humidity had dropped to roughly 52 to 59 percent in the afternoon, cloud cover was only about 42 percent near the event, and boundary-layer height was elevated near 1200 m after much deeper afternoon mixing earlier in the day.

Mark exactly one:

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Note (optional):

Claim 15 of 35

The anomaly occurred on July 4th, 2026, at approximately 23:01:55+00:00.

Mark exactly one:

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Note (optional):

Claim 16 of 35

Eastern Houston transport at 2026-07-04 23:00 UTC was consistent with delivery of an aged urban-industrial plume because surface winds near the anomaly were from roughly 160 to 170 degrees at about 5 to 8 m/s.

Mark exactly one:

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Note (optional):

Claim 17 of 35

A monitor-scale local enhancement or measurement artifact is a plausible contributor because the openaq ozone peak of 0.101 ppm at sensor 2039 was highly isolated relative to the surrounding network and was not accompanied by elevated nearby NO₂, CO, SO₂, PM_{2.5}, or PM₁₀.

Mark exactly one:

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Note (optional):

Claim 18 of 35

The ozone anomaly at location (29.768, -95.221) at 2026-07-04T23:00:00+00:00 had an expected value of 0.01856291390728477 ppm and a z-score of 6.554339250396881, and the anomaly severity was classified as severe.

Mark exactly one:

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Note (optional):

Claim 19 of 35

Peak afternoon air temperatures in the area reached between 33.9°C and 35.0°C on July 4, 2026, facilitating rapid photochemical production of ground-level ozone.

Mark exactly one:

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Note (optional):

Claim 20 of 35

Nitrogen dioxide precursor emissions provided the necessary baseline reactants for photochemically generated ozone accumulation across the Houston industrial and urban corridor.

Mark exactly one:

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Note (optional):

Claim 21 of 35

The air quality anomaly in Houston, Texas is likely caused by a combination of factors including high levels of PM2.5 and NO2 emissions from industrial activities and vehicle traffic.

Mark exactly one:

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Note (optional):

Claim 22 of 35

The air quality anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter and volatile organic compounds.

Mark exactly one:

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Note (optional):

Claim 23 of 35

The ozone anomaly is most consistent with a localized aged photochemical plume rather than fresh combustion, smoke, or a sulfur-rich industrial release because ozone reached 0.101 ppm while nearby co-pollutants remained low, including NO2 near 1.7 ppb, CO near 0.2 ppm, SO2 near 0.0 ppb, PM2.5 near 6 to 7 ug/m3, PM10 near 24 ug/m3, and satellite SO2 and CO columns without unusual enhancement.

Mark exactly one:

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Note (optional):

Claim 24 of 35

An ozone concentration spike of 0.101 ppm was detected at coordinates (29.768, -95.221) on 2026-07-04 at 23:00:00 UTC, substantially exceeding the expected baseline of approximately 0.0186 ppm with a z-score of 6.55.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 25 of 35

A monitor-scale local enhancement or measurement artifact remains plausible because the openaq ozone peak of 0.101 ppm at the anomalous site was far above the network mean for the same 6-hour period while nearby co-pollutants were not concurrently elevated, with TCEQ NO₂ near 1.7 ppb, TCEQ CO near 0.2 ppm, TCEQ SO₂ near 0.0 ppb, and PM_{2.5} remaining low at nearby openaq and purpleair sensors.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 26 of 35

Transport of an aged urban-industrial plume into eastern Houston at the anomaly time is supported by southerly to south-southeasterly flow of about 5 to 8 m/s near the event, which is consistent with advection from the Houston Ship Channel and broader urban core toward the eastern Houston receptor.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 27 of 35

The air quality anomaly in Houston, Texas is caused by high levels of SO₂ particles.

Mark exactly one:

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Note (optional):

Claim 28 of 35

The temperature and wind speed data suggest that the air quality anomaly was exacerbated by a heatwave and stagnant atmospheric conditions.

Mark exactly one:

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Note (optional):

Claim 29 of 35

Houston-area meteorology at 2026-07-04 23:00 UTC favored strong ozone production and persistence because afternoon temperature was about 33 C, relative humidity was only about 59 to 70 percent near the anomaly, cloud cover was about 42 percent, and boundary-layer height was about 1200 m near the event time.

Mark exactly one:

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Note (optional):

Claim 30 of 35

The air quality anomaly in Houston, Texas is caused by high levels of NO₂ particles.

Mark exactly one:

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Note (optional):

Claim 31 of 35

The anomaly is due to a combination of factors including high temperatures, humidity, and stagnant atmospheric conditions, which have led to the formation of ground-level ozone and other pollutants.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 32 of 35

The CO, NO₂, and SO₂ concentrations were significantly higher than normal levels during the 6-hour period preceding the anomaly, indicating a potential source of pollution.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 33 of 35

The pollutant showing the anomaly is ozone, measured at 0.101 ppm at 2026-07-04T23:00:00+00:00 at location (29.768, -95.221).

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 34 of 35

High surface temperatures exceeding 33 °C during the afternoon of July 4 enhanced photochemical reaction rates that drive daytime ground-level ozone formation in Houston.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 35 of 35

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
